S.A. ENGINEERING COLLEGE- CHENNAI-77

(Autonomous- Institute Level Research Centre, Affiliated to Anna University)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

AD1801A PROJECT PHASE - II

BATCH NUMBER : AD33

AI Automated Invoice Lifecycle Management

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- Sustainable Development Goals
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- Literature Survey
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- Problem Statement
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Scope & Motivation

- To develop an AI-based system that automatically reads, extracts, and validates invoice data
- It supports multiple invoice formats, and seamlessly integrates with CRM and ERP platforms.
- Reduce human errors and operational delays in invoice handling.
- It ensures the accuracy, security, scalability, and smooth business workflow automation.

- The challenges faced in manual invoice processing, such as slow operations, human errors, data mismatch between CRM and ERP systems.
- Existing tools only solve isolated parts of the workflow.
- It delays with payments, and lack of financial visibility.
- It creates a strong need for an intelligent, automated, accurate, and efficient system that supports digital transformation.

Sustainable Development Goals

SDG 8 – Decent Work and Economic Growth

The system reduces repetitive manual work, enabling employees to focus on high-value tasks such as analysis and decision-making. It enhances operational efficiency and supports sustainable economic growth by automating financial processes.

SDG 9 – Industry, Innovation, and Infrastructure

The project directly contributes to building smart digital infrastructure within organizations. By integrating AI with CRM and ERP systems, it promotes innovation, modernizes enterprise operations, and strengthens technological capabilities.

SDG 12 – Responsible Consumption and Production

By Digitizing and automating Invoices, The System reduces paper usage and supports sustainable business partices.

SDG 16 – Peace, Justice and Strong Institutions

Automated invoice validation, fraud detection, audit trails, and transparent financial workflows reduce corruption risks, strengthen internal controls, and promote accountability in organizations.

Introduction

This system uses AI and Vision Language Models to automate the entire invoice lifecycle by extracting structured data from PDFs and images with high precision. It features a CRM-driven recommendation engine that analyzes patterns to suggest products, alongside automated classification and approval tracking. The platform integrates seamlessly with ERP and inventory modules, ensuring real-time stock updates and synchronized financial workflows. A centralized dashboard provides graphical insights into trends and stock levels, supported by audit logs for enterprise compliance. Ultimately, this project transforms manual administrative tasks into a streamlined, data-driven operation that enhances overall business efficiency.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
1.	AI POWERED INVOICE PROCESSING SYSTEM	2025	Journal: International Journal of Research Publication and Reviews ISSN 2582-7421,	The AI Automated Invoice Lifecycle Management system significantly improves financial workflow accuracy, speed, and transparency by automating invoice processing from ingestion to reporting.
2	AI-Powered Invoice Automation in ERP Systems: Revolutionizing Accounts Payable	2025	Journal: Journal of Computer Science and Technology Studies, ISSN: 2709-104X DOI: 10.32996/jcsts	Integrating AI with cloud-based ERP systems revolutionizes accounts payable by enabling faster, more accurate, and cost-efficient invoice processing. Advanced analytics and automation deliver strong ROI, improved cash flow visibility, and better vendor management

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
3.	Intelligent Document Processing: Automating Invoice and Contract Management with AI	2024	Journal: Research Gate13(3):892-897,DOI:10.21275/SR24313021820)	AI-based IDP using OCR, NLP, and ML significantly enhances efficiency, accuracy, and compliance in invoice and contract processing.
4.	FinTech - Automatic Payment Process in the ERP System	2024	Journal: International Journal of Computer Trends and Technology (IJCTT72(1):99-103, DOI: 10.14445/22312803/IJCTT-V72I11P116	ERP with AI, cloud, and blockchain automates finance, boosts security, and scalability, with triple-entry accounting aiding fraud prevention.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
5.	Enhancing Customer Relationship Management in ERP Systems through AI: Personalized Interactions, Predictive Modeling, and Service Automation	2024	Journal: International Journal of Science and Research (IJSR) 13(3):892-897,DOI:10.21275/SR24313021820)	AI-integrated ERP-CRM boosts personalization, efficiency, and real-time decision-making, addressing Indian market gaps while tackling privacy and ethical concerns.
6.	An Introduction to Vision-Language Modeling	2024	Journal: Research Gate , arXiv:2405.17247v1	Vision-Language Modeling (VLM) represents a critical leap in artificial intelligence by integrating visual perception with linguistic reasoning into a single multimodal framework, thereby enabling machines to interpret complex, unstructured data through a shared semantic understanding.

Literature Survey

S. no	Title of the Paper	Year	Journal/ Conference	Inference
7.	AI Enabled Invoice Management Application	2023	Journal: 8th International Conference on Communication and Electronics Systems (ICCES) | 979-8-3503-9663-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICCES57224.2023.10192802	AI-enabled invoice management automates data extraction, validation, and approval, reducing errors and processing time for SMEs. The system improves efficiency and fraud detection, though cost and standardization remain key challenges.
8.	An Overview of Data Extraction from Invoices	2022	Journal: IEEE Access PP(99):1-1 DOI:10.1109/ACCESS.2024.3360528	OCR, NLP, and graph-based ML extract invoice data despite layout challenges, enabling adaptable, low-intervention document management.

Objectives

- It Uses OCR and VLM technologies to retrieve data from PDF and image invoices.
- It can Transforms raw text into organized fields like vendor details, line items, and totals.
- It Ensures data accuracy and completeness while proactively identifying duplicate invoices.
- It can Categorizes invoices by rules and monitors their progress through every approval stage.
- It Synchronizes information with CRM and ERP modules to automate inventory and finances.
- It Provides data visualization through dashboards and audit logs for full transparency

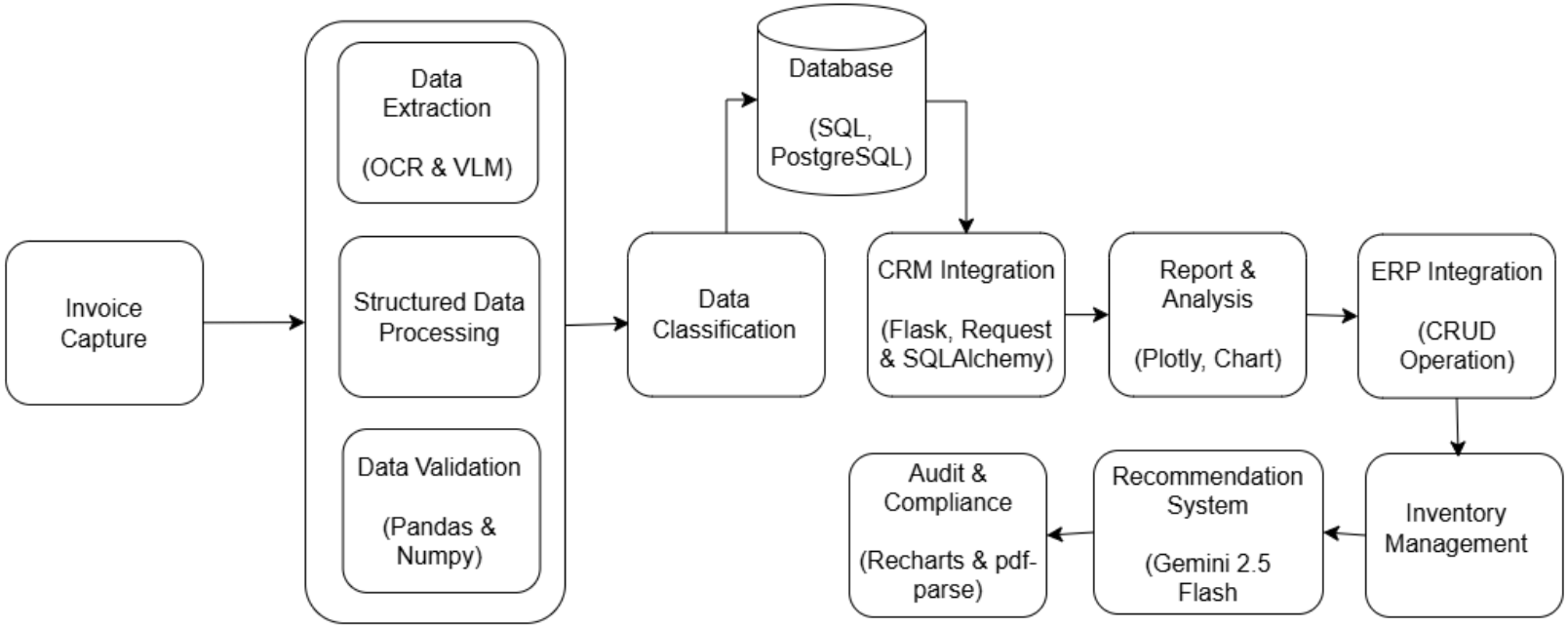
Problem Statement

- Inefficient Manual Workflows: Organizations rely on slow, labor-intensive manual entry for diverse invoice formats, leading to significant delays and high operational costs.

- Data Integrity and Oversight Risks: The lack of automated validation tools results in frequent human errors—such as incorrect entries or missing details—and makes tracking an invoice's progress through approval stages difficult.

- System Fragmentation: The inability to integrate invoice data with CRM, ERP, and inventory systems prevents real-time synchronization, hindering accurate financial forecasting and effective decision-making.

Proposed Work-Block Diagram



Proposed-Novel Idea

Design an AI-based web application that seamlessly automates the entire invoice processing:

- AI Powered system to automate the entire invoice lifecycle
- Intelligent data extraction using OCR + VLM
- Real-Time validation with CRM & ERP integration
- Inventory Management with CRM & ERP
- Automatic Approval workflow and status tracking

Proposed-Modules

- Invoice Capture
- Data Extraction
- Structured Data Processing
- Data Validation
- Data Classification
- CRM Integration
- Report & Analysis
- ERP Integration
- Inventory Management
- Recommendation System
- Audit & Compliance

Proposed-Module Description

1. **Data Ingestion**

- Collects invoices from multiple sources: PDF, document uploads, scanned copies, and Images.
- Uses OCR (Optical Character Recognition) to extract invoice details such as invoice number, date, vendor name, line items, quantity, unit price, and total amount.
- Converts unstructured documents into a machine-readable format for downstream processing.

2. **AI-Powered Data Extraction (VLM & OCR)**

- The Goal: Uses Gemini 1.5 Flash (Vision Language Model) to "read" invoices like a human, understanding both text and spatial layout.
- The Benefit: Converts unstructured PDFs and images into precise JSON data, eliminating the 3%–5% error rate of manual entry.
- Key Tech: VLM for semantic understanding and OCR for character recognition.

Proposed-Module Description

3. <u>Intelligent Validation & Classification</u>

- The Goal: Automatically verifies that all mandatory fields (tax, total, date) are present and cross-checks them for accuracy.
- The Benefit: Proactively identifies duplicate invoices and categorizes them by department or priority for faster processing.
- Key Tech: Rule-based logic and RAG (Retrieval-Augmented Generation) for data verification.

4. <u>CRM-Driven Recommendation Engine</u>

- The Goal: Analyzes historical invoice patterns to identify customer preferences and vendor performance.
- The Benefit: Recommends relevant products or services based on past interactions, turning administrative data into a sales and growth tool.
- Key Tech: Predictive modeling and customer sentiment analysis.

Proposed-Module Description

5. <u>ERP & Financial Workflow Integration</u>

- The Goal: Simulates a seamless connection between the invoice system and enterprise financial modules
- .The Benefit: Synchronizes approved invoices with financial records automatically, ensuring real-time visibility into cash flow and payables.
- Key Tech: API-driven synchronization and automated financial status tracking.

6. <u>Inventory Management Module</u>

- The Goal: Automatically updates stock levels based on the line items and quantities extracted from validated invoices.
- The Benefit: Prevents stockouts and eliminates the need for manual inventory logging, maintaining an accurate "Order-to-Cash" cycle.
- Key Tech: Automated stock increment logic ($New_Stock = Current + Invoiced$).

Proposed-Module Description

7. **Report Analysis Module**

- It generated the dashboard, summaries and performance insights to support decision making and financial planning.

8. Analytical Dashboard & Audit Logs

- The Goal: A centralized hub featuring graphical charts for stock trends, processing statistics, and transaction summaries.

- The Benefit: Provides stakeholders with actionable insights while maintaining a 100% transparent audit trail for enterprise compliance.

- Key Tech: React.js data visualization and immutable activity logging.

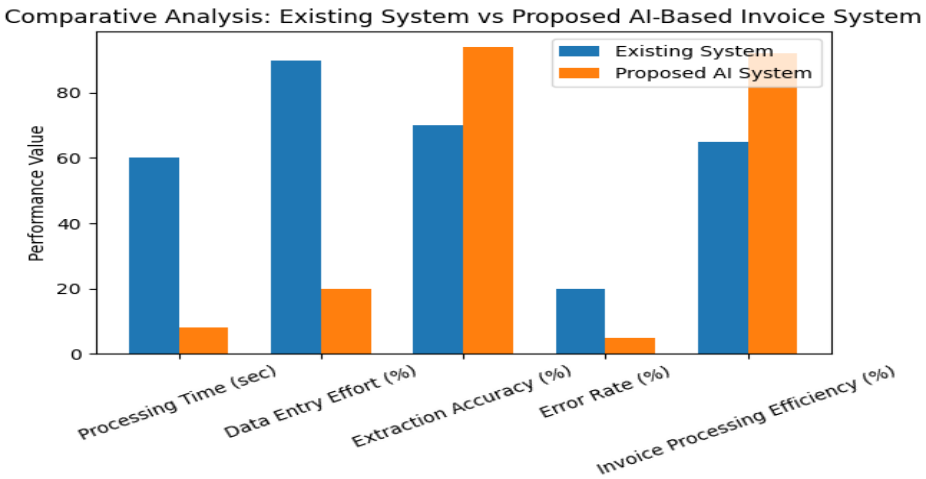
Result

The developed system successfully automates invoice processing and improves operational efficiency.

Key results include:

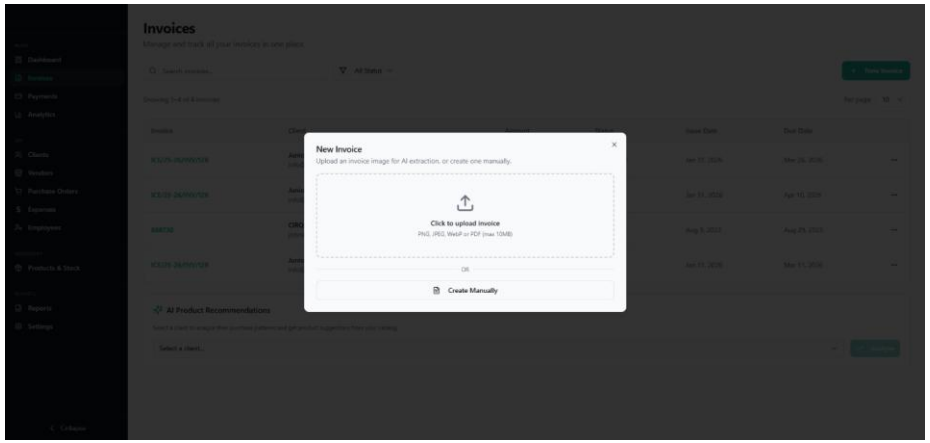
- Faster invoice processing
- Higher data extraction accuracy
- Reduced manual data entry
- Improved invoice tracking
- Automated reporting and insights

The system demonstrates how AI can significantly improve business workflow automation.

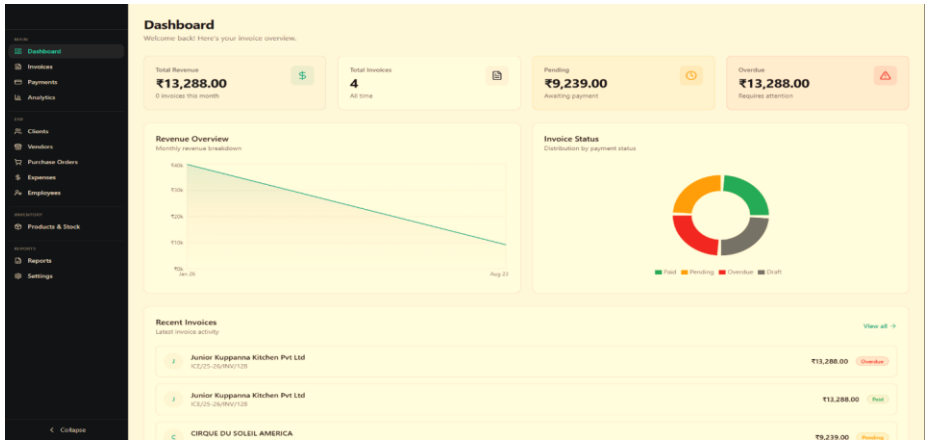


Snapshots

1. Upload Invoice:

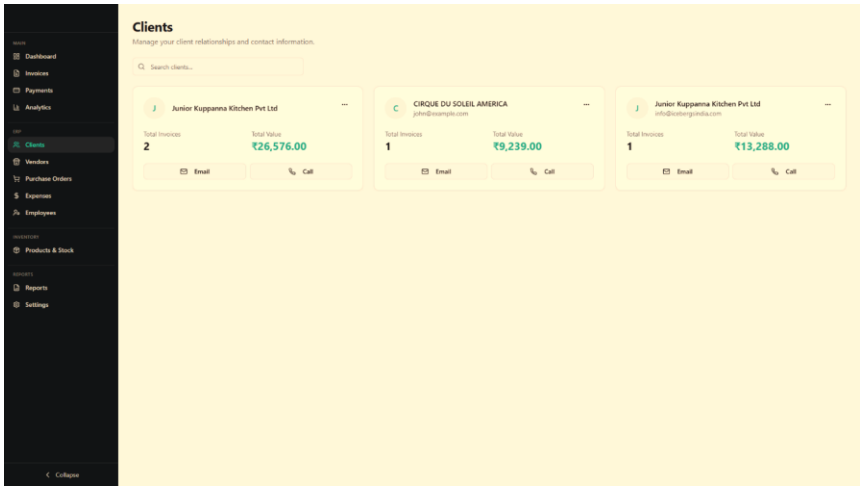


2. Invoice Dashboard:

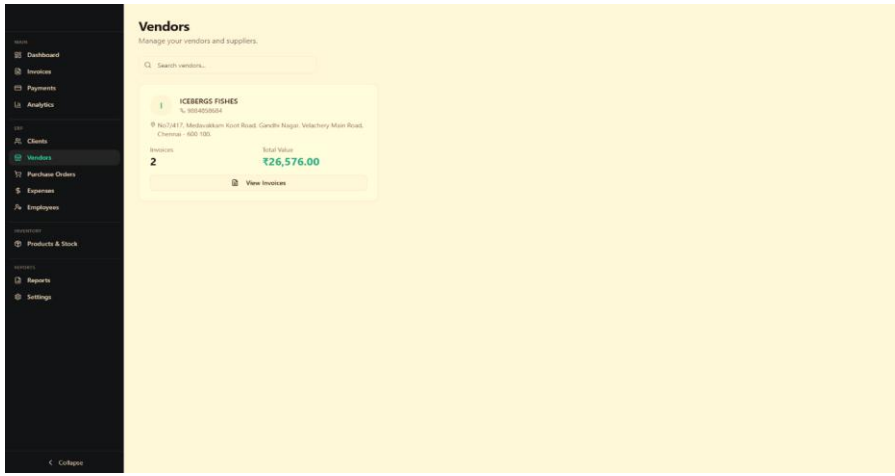


Snapshots

3. Customer Dashboard:

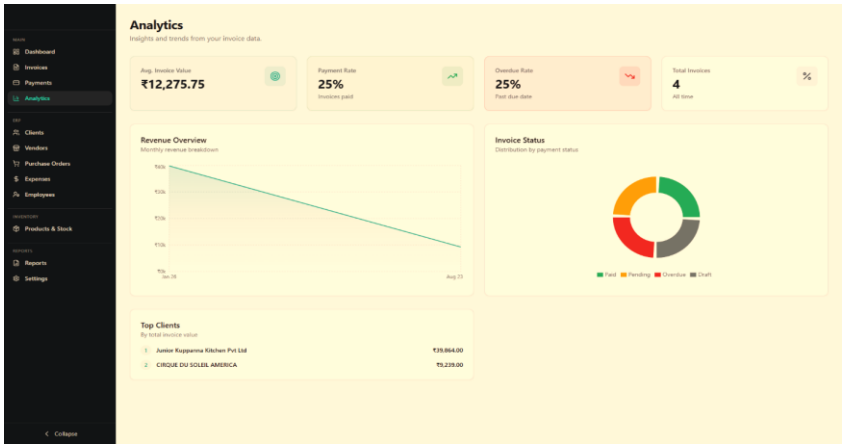


4. Vendor Dashboard:

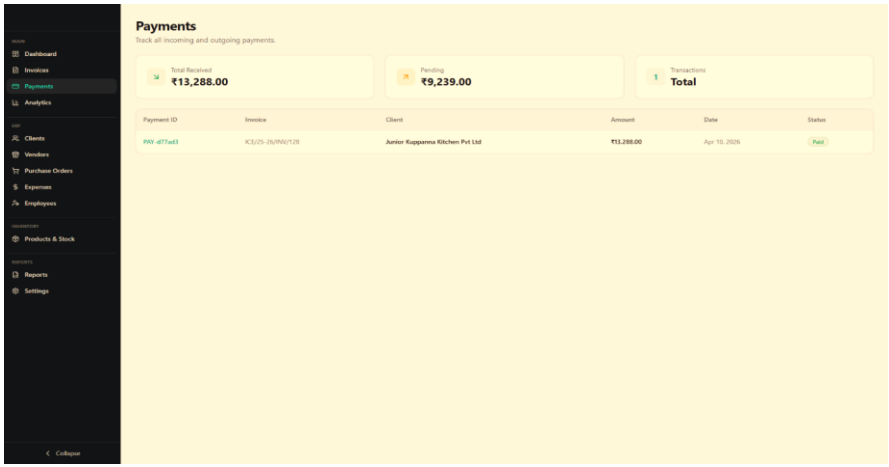


Snapshots

5. Analysis Dashboard:

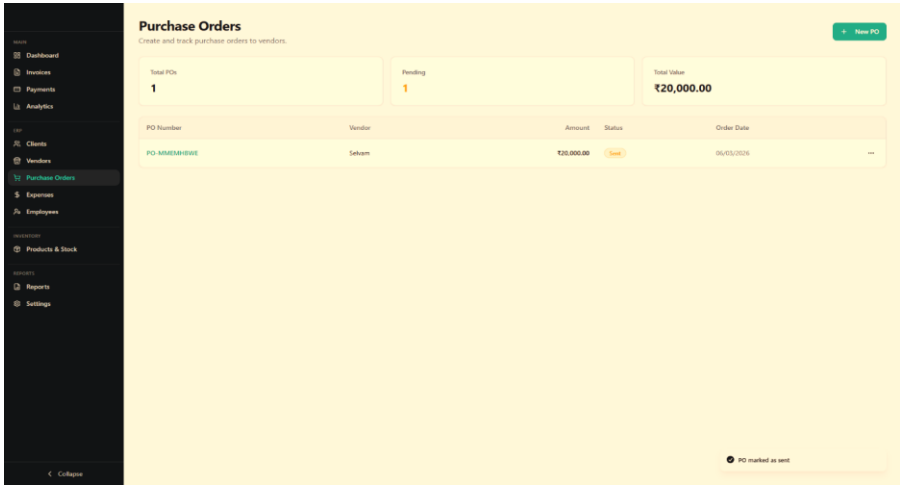


6. Payment Dashboard:

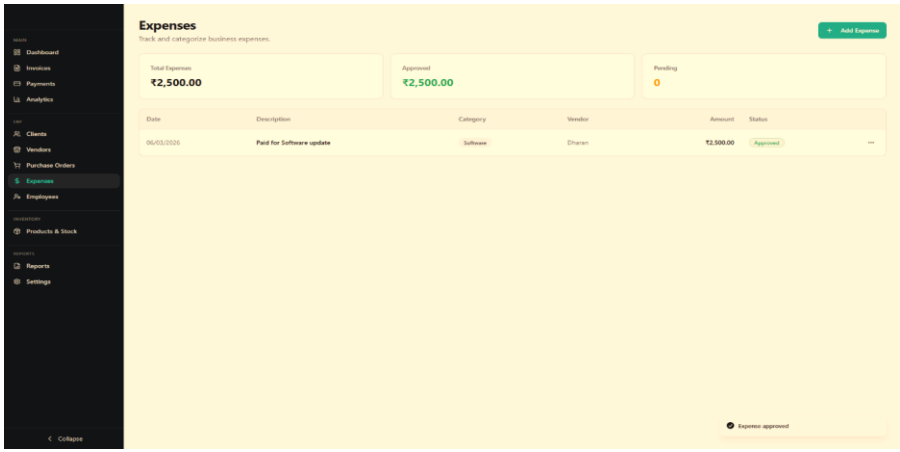


Snapshots

7. Purchase Order Dashboard:

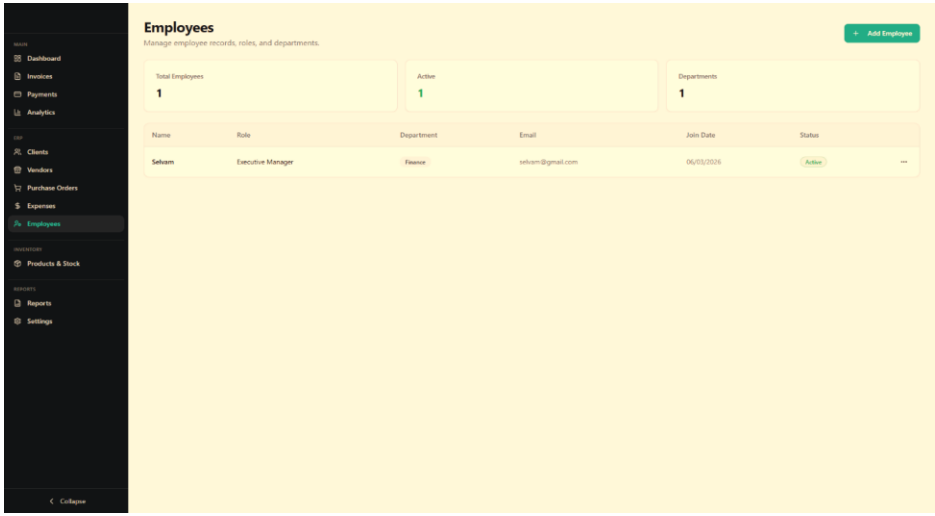


8. Expenses Dashboard:

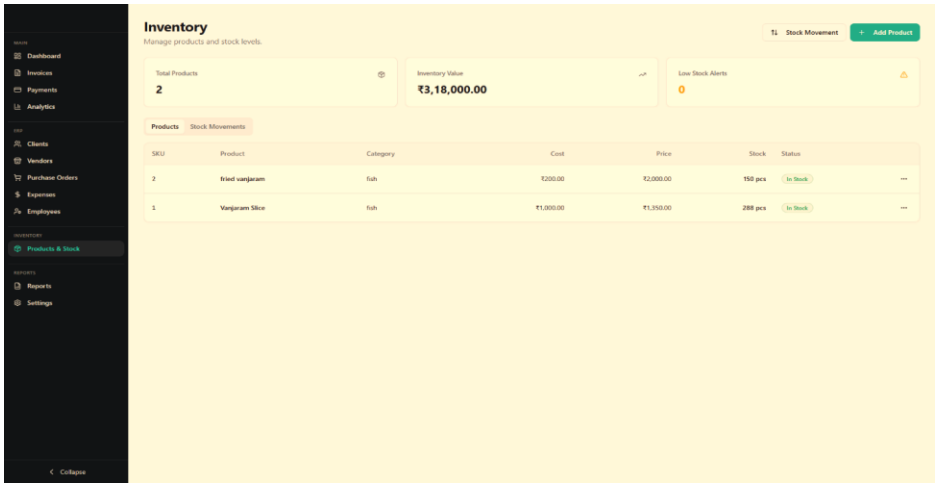


Snapshots

9. Employees Dashboard:



10. Inventory Management Dashboard:



Conclusion

- The AI-Based Automated Invoice Lifecycle Management System provides an efficient solution for automating invoice processing. By using technologies such as OCR and Vision Language Models (VLM), the system can automatically extract and organize important information from invoice documents.

- The system reduces manual data entry, processing time, and human errors, while improving the accuracy and efficiency of invoice management. Integration with CRM, ERP, and inventory modules helps streamline business workflows and ensures better data management.

- Overall, the project demonstrates how artificial intelligence can be effectively applied to automate financial document processing and improve organizational productivity.

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[2] Koti Reddy Onteddu, "AI-Powered Invoice Automation in ERP Systems: Revolutionizing Accounts Payable", Journal of Computer Science and Technology Studies ISSN: 2709-104X DOI: 10.32996/jcsts , May 2025.

[3] Bryson Santos, "Intelligent Document Processing: Automating Invoice and Contract Management with AI", Research Gate, October 2024.

[4] Srinivasa Rao Gunturu, Madhavi Godbole, Hari Prasad Josyula, "FinTech-Automatic Payment Process in the ERP System", International Journal of Computer Trends and Technology 72(1):99-103, DOI: 10.14445/22312803/IJCTT-V72I11P116, February 2024.

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